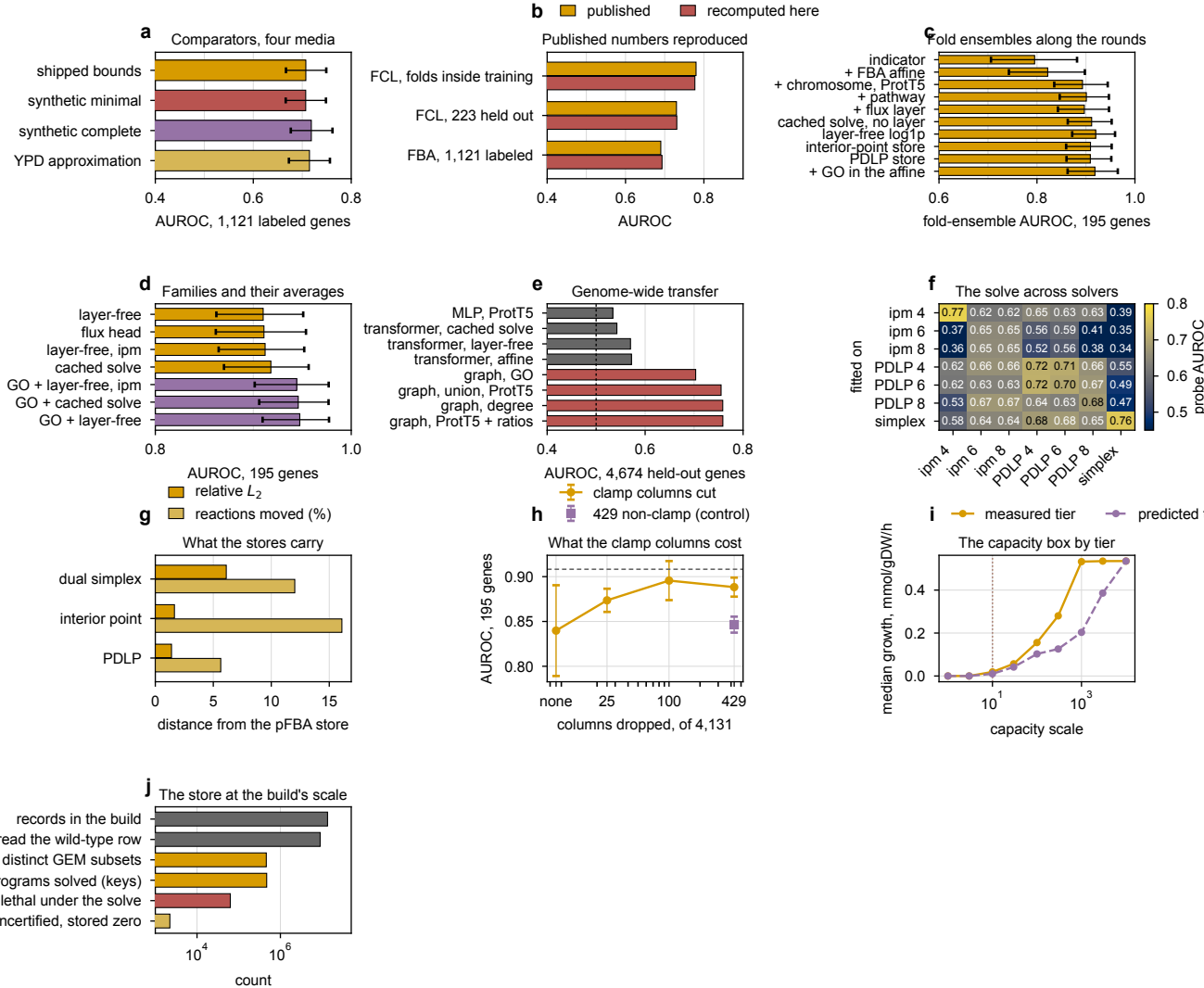
a
Comparators, four media
shipped bounds
synthetic minimal
synthetic complete
YPD approximation
0.4
0.6
0.8
AUROC, 1,121 labeled genes
b
published
recomputed here
Published numbers reproduced
FCL, folds inside training
FCL, 223 held out
FBA, 1,121 labeled
0.4
0.6
0.8
AUROC
c
Fold ensembles along the rounds
indicator
+ FBA affine
+ chromosome, ProtT5
+ pathway
+ flux layer
cached solve, no layer
layer-free log1p
interior-point store
PDLP store
+ GO in the affine
0.6
0.8
1.0
fold-ensemble AUROC, 195 genes
d
Families and their averages
layer-free
flux head
layer-free, ipm
cached solve
GO + layer-free, ipm
GO + cached solve
GO + layer-free
0.8
1.0
AUROC, 195 genes
relative L_2
reactions moved (%)
e
Genome-wide transfer
MLP, ProtT5
transformer, cached solve
transformer, layer-free
transformer, affine
graph, GO
graph, union, ProtT5
graph, degree
graph, ProtT5 + ratios
0.4
0.6
0.8
AUROC, 4,674 held-out genes
clamp columns cut
429 non-clamp (control)
f
The solve across solvers
fitted on
ipm 4 0.77 0.62 0.62 0.65 0.63 0.63 0.39
ipm 6 0.37 0.65 0.65 0.56 0.59 0.41 0.35
ipm 8 0.36 0.65 0.65 0.52 0.56 0.38 0.34
PDLP 4 0.62 0.66 0.66 0.72 0.71 0.66 0.55
PDLP 6 0.62 0.63 0.63 0.72 0.70 0.67 0.49
PDLP 8 0.53 0.67 0.67 0.64 0.63 0.68 0.47
simplex 0.58 0.64 0.64 0.68 0.68 0.65 0.76
ipm 4
ipm 6
ipm 8
PDLP 4
PDLP 6
PDLP 8
simplex
0.8
0.7
0.6
0.5
probe AUROC
g
What the stores carry
dual simplex
interior point
PDLP
0
5
10
15
distance from the pFBA store
h
What the clamp columns cost
AUROC, 195 genes
0.90
0.85
0.80
none
25
100
429
columns dropped, of 4,131
i
measured tier
predicted
The capacity box by tier
median growth, mmol/gDW/h
0.4
0.2
0.0
10^1
10^3
capacity scale
j
The store at the build's scale
records in the build
read the wild-type row
distinct GEM subsets
programs solved (keys)
lethal under the solve
uncertified, stored zero
10^4
10^6
count